

JEE Main - 1 | JEE 2023

Date: 30/03/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks are **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 9 (both inclusive).
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

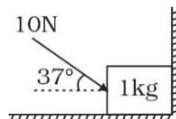
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

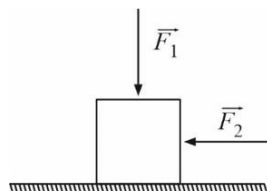
PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

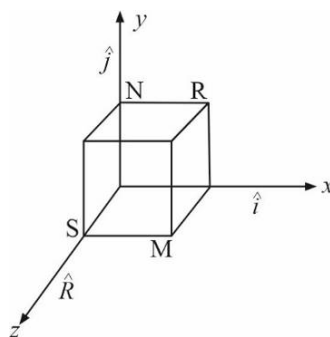
1. A force of 10N is applied as shown on a 1 kg block which is kept on a horizontal ground. The block is held at rest against a vertical wall. The normal reaction by the ground and the normal reaction by the wall will respectively be: (Take $g = 10 \text{ ms}^{-2}$ & $\cos 37^\circ = \frac{4}{5}$)



- (A) $5\sqrt{2}N, 5\sqrt{2}N$ (B) $10\sqrt{2}N, 10\sqrt{2}N$
 (C) $16N, 8N$ (D) $8N, 16N$
2. A block of mass 5 kg is at rest on a sufficiently rough surface. A vertical force $F_1 = 30N$ and a horizontal force $F_2 = 40N$ is acting on it (as shown in the figure). What will be the magnitude of net force applied by the ground on the block: ($g = 10 \text{ m/s}^2$)

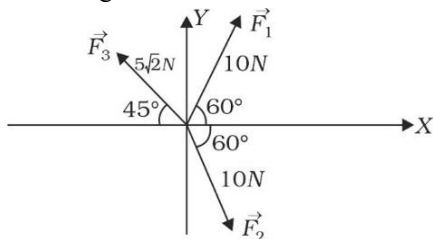


- (A) $40\sqrt{5}N$ (B) $50N$ (C) $80N$ (D) $50\sqrt{2}N$
3. For the given cube of side length $2m$, the unit vector in the direction of $\overrightarrow{MN} + \overrightarrow{RS}$ will be:

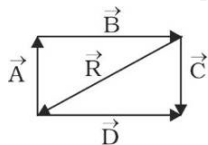


- (A) $\frac{-\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$ (B) $-\hat{j}$ (C) $-\hat{k}$ (D) $-\hat{i}$
4. The component of vector $\vec{V} = 3\hat{i} + 2\hat{j}$ along the vector $\vec{R} = 3\hat{i} - 4\hat{j}$ will be :
- (A) 1 (B) $\frac{1}{5}$ (C) 17 (D) 5
5. Which of the sets given below may represent the magnitudes of four vectors adding to zero?
- (A) 2, 3, 7, 1 (B) 5, 9, 4, 2 (C) 1, 1, 1, 4 (D) 13, 2, 9, 1

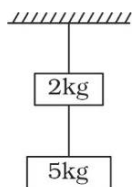
6. The magnitude of the resultant of the given forces is:



- (A) 5 N (B) $5\sqrt{2}$ N (C) 10 N (D) $10\sqrt{2}$ N
7. Work done by the force $\vec{F} = (3\hat{i} - 4\hat{j} + 5\hat{k})$ N, for a displacement of $\vec{r} = (2\hat{i} + 4\hat{j} + 2\hat{k})$ metre, will be :
 (A) 32 Joule (B) 16 Joule (C) 8 Joule (D) 0 Joule
8. Magnitudes of two vectors $\vec{P}$ and $\vec{Q}$ are 3 units and 4 units respectively. If angle between $\vec{P}$ and $\vec{Q}$ is 120° , then the magnitude of the resultant of $\vec{P}$ and $\vec{Q}$ will be:
 (A) $\sqrt{10}$ units (B) $\sqrt{11}$ units (C) $\sqrt{12}$ units (D) $\sqrt{13}$ units
9. If $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = 6\hat{i} + C\hat{j} + 3\hat{k}$, find the value of C so that vectors $\vec{a}$ and $\vec{b}$ become parallel.
 (A) -2 (B) 3 (C) 7.5 (D) -6
10. The (x, y, z) co-ordinates of two points P_1 and P_2 are given respectively as $(2, 3, 4)$ and $(1, 5, 2)$. The displacement vector from P_2 to P_1 is given by :
 (A) $-i + 2\hat{j} - 2\hat{k}$ (B) $3i + 8\hat{j} + 6\hat{k}$ (C) $i - 2\hat{j} + 2\hat{k}$ (D) $i + 2\hat{j} + 2\hat{k}$
11. Vector $\vec{P}$ and vector $\vec{Q}$ have their magnitudes in the ratio of 3 : 1 respectively. The magnitude of $(\vec{P} - \vec{Q})$ is 2 times the magnitude of $(\vec{P} + \vec{Q})$. The angle between $\vec{P}$ and $\vec{Q}$ is:
 (A) $\cos^{-1}\left(-\frac{1}{4}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}(-1)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$
12. In the given figure, vector $\vec{R}$ is equal to :

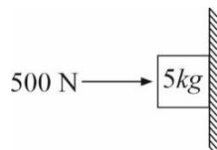


- (A) $\vec{C} + \vec{D}$ (B) $\vec{B} + \vec{C}$ (C) $\vec{C} - \vec{D}$ (D) $\vec{A} - \vec{B}$
13. In the given arrangement, what will be the value of tension in the string connecting the two blocks:
 (Take $g = 10 \text{ m/s}^2$. The system is in equilibrium.)

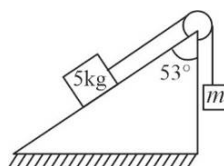


- (A) 70 N (B) 20 N (C) 50 N (D) 30 N
14. If $|\vec{P}| = 5$, $|\vec{Q}| = 12$ and $|\vec{P} + \vec{Q}| = 13$, find the value of $(2\vec{P} + \vec{Q}) \cdot (\vec{P} - 2\vec{Q})$.
 (A) 410 (B) -410 (C) 238 (D) -238
15. The angle between vector $\vec{E} = 3\hat{i} + 2\hat{j}$ and vector $\vec{r} = 2\hat{i} - 3\hat{j}$ will be:
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) 0 (D) $\frac{\pi}{2}$

16. The unit vector perpendicular to vector $\vec{r} = 2\hat{i} + 3\hat{j} - \hat{k}$ and vector $\vec{p} = \hat{i} + \hat{j} + 2\hat{k}$ will be:
- (A) $\frac{7\hat{i} - 5\hat{j} - \hat{k}}{5\sqrt{3}}$ (B) $\frac{7\hat{i} + 5\hat{j} - \hat{k}}{5\sqrt{3}}$ (C) $\frac{2\hat{i} + 3\hat{j} - 2\hat{k}}{\sqrt{17}}$ (D) $\frac{2\hat{i} - 3\hat{j} - 2\hat{k}}{\sqrt{17}}$
17. A block of mass 5kg is pressed against a wall as shown. If coefficient of static friction between the block and wall is 0.2 and the block is at rest, the magnitude of friction acting between the block and wall is:



- (A) 100 N (B) 50 N (C) 25 N (D) None of these
18. If $|\vec{A} \times \vec{B}| = \frac{1}{\sqrt{3}} |\vec{A} \cdot \vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ will be :
- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) zero
19. The blocks are at rest on a smooth and fixed wedge. The reaction force of wedge over the pulley has a magnitude of : (Take $g = 10\text{m/s}^2$ & $\cos 37^\circ = \frac{4}{5}$)



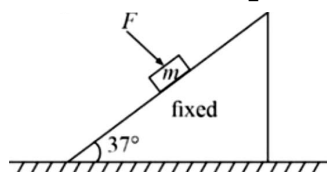
- (A) 120 N (B) $\frac{120}{\sqrt{5}}\text{N}$ (C) $30\sqrt{2}\text{N}$ (D) None of these
20. A vector $\vec{R}$ is in y-z plane. If z-component of $\vec{R}$ is 5 unit and y-component is $-5\sqrt{3}$ units, then the angle made by $\vec{R}$ with the z-axis will be:
- (A) 30° (B) 60° (C) 120° (D) 150°

SPACE FOR ROUGH WORK

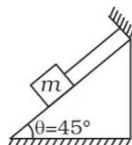
SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. The coefficient of friction between a block of mass 10kg and inclined plane on which it is kept is $\mu = 0.6$. If the minimum force F (in newton) with which it should be pressed normally against the incline to keep it at rest is $10y$, then y is _____. [Take $g = 10\text{ m/s}^2$ & $\tan 37^\circ = \frac{3}{4}$]



22. For what value of m , the vector $\vec{E} = 3\hat{i} + m\hat{j} - 2\hat{k}$ and vector $\vec{B} = m\hat{i} + 7\hat{j} + 5\hat{k}$ are orthogonal?
23. A block of mass $m = 10\sqrt{2}\text{ kg}$ is tied to a string and held on an inclined plane of inclination 45° as shown in the figure. If the normal reaction is $25x$ (in Newton) between the block and the incline plane, then x is _____. (Take $g = 10\text{ m/s}^2$.)



24. If vectors $\vec{m} = (2\hat{i} - 3\hat{j})m$ and $\vec{n} = (-5\hat{i} + 7\hat{j})m$ represent two adjacent sides of a parallelogram, then the area of the parallelogram will be _____ (in m^2).
25. If $\vec{a} = 3\hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{b} = 6\hat{i} + 23\hat{j} + 5\hat{k}$, then the value of $\frac{\vec{a} \cdot \vec{b}}{13}$ will be _____.

SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- 10 g of CaCO_3 reacted with 19.6 g of H_2SO_4 and 6g CaCO_3 remained unutilized. What was the percent purity of H_2SO_4 ? (Mo of Ca = 40, C = 12, O = 16, H = 1, S = 32)
 (A) 40 (B) 20 (C) 10 (D) 5
- Calculate volume (in mL) of 0.08 M NaOH needed to titrate 50 ml of 0.06 M H_3PO_4 .
 (A) 37.5 mL (B) 112.5 mL (C) 75 mL (D) 100 mL
- For the neutralization reaction, $\text{Ba}(\text{OH})_2 + 2\text{HCl} \longrightarrow \text{BaCl}_2 + 2\text{H}_2\text{O}$
 How much amount (approx. in gms) of $\text{Ba}(\text{OH})_2$ is required to neutralize 300 ml of 0.4 M HCl?
 (Ba = 137.3, O = 16, H = 1, Cl = 35.5)
 (A) 10 g (B) 60 g (C) 1027 g (D) 20 g
- According to the equation, $4\text{NH}_3 + 5\text{O}_2 \longrightarrow 4\text{NO} + 6\text{H}_2\text{O}$ when 1 mole of O_2 and 1 mole of ammonia are mixed, then:
 (I) 1.5 mole of H_2O is produced
 (II) 0.8 mole of NO is produced
 (III) all O_2 will be consumed
 (IV) all NH_3 will be consumed
 (A) I and III (B) I and IV (C) II and III (D) II and IV
- 2N H_2SO_4 will have same molar concentration as :
 (A) 2M H_3PO_2 (B) 4M H_3PO_2 (C) 1M H_3PO_2 (D) 2N H_3PO_2
- 20 ml of 0.1M AlCl_3 is mixed with 30 ml of 0.3M NaOH. What is the mass (in gm) of $\text{Al}(\text{OH})_3$ formed?
 (Mo $\text{Al}(\text{OH})_3 = 78 \text{ g}$)
 (A) 0.512 (B) 0.156 (C) 156 (D) 0.234
- 0.2g mixture of NaOH, Na_2CO_3 and some inert impurities was first titrated with $\frac{N}{10}$ HCl using phenolphthalein as an indicator, 17.5 ml of HCl was required at the end point. After this methyl orange was added and titrated, 1.5 ml of same HCl was required for next end point. The weight percentage of Na_2CO_3 in the mixture is: (Mo of Na = 23, C = 12, O = 16)
 (A) 4 (B) 8 (C) 10 (D) 20
- From the following reaction sequence,
 $\text{Cl}_2 + 2\text{KOH} \longrightarrow \text{KCl} + \text{KClO} + \text{H}_2\text{O}$ 50% yield
 $3\text{KClO} \longrightarrow 2\text{KCl} + \text{KClO}_3$ 75% yield
 $4\text{KClO}_3 \longrightarrow 3\text{KClO}_4 + \text{KCl}$ 100% yield
 Calculate the mass of chlorine needed to produce 13.85 g KClO_4 (Mo. of $\text{KClO}_4 = 138.5 \text{ g}$).
 (A) 75.2 g (B) 378.6 g (C) 28.4 g (D) 54.6 g
- Mole fraction of methanol (CH_3OH) in aqueous solution is 0.2. Molality of solution will be:
 (A) 25 (B) 13.8 (C) 10 (D) 4

10. Rearrange the following (I to IV) in the order of increasing masses:
 (I) 0.5 mole of O_2
 (II) 0.5 gm atom of oxygen
 (III) 6.02×10^{23} molecules of O_2
 (A) $I < II < III$ (B) $III < II < I$ (C) $II < I < III$ (D) $II < III < I$
11. 1 M HCl and 2M HCl are mixed in volume ratio of 5 : 1. What is the final molarity of HCl solution?
 (A) 1.1 (B) 1.5 (C) 2M (D) 3M
12. 2g of solid dibasic acid is completely neutralized by 25 ml of 0.5M $Ba(OH)_2$ solution. Molecular mass of Acid is:
 (A) 150 g/mol (B) 250 g/mol (C) 500 g/mol (D) 160 g/mol
13. What volume of 25% alcohol by weight ($d = 0.5\text{g/ml}$) must be used to prepare 200 ml of 10% alcohol by mass ($d = 1.5\text{ g/ml}$)?
 (A) 200 ml (B) 240 ml (C) 120 ml (D) 80 ml
14. Select the dimensionless quantity.
 (I) Molarity (II) Mass fraction (III) Density (IV) Mole Fraction
 (A) I, II, IV (B) II and IV (C) III and IV (D) only IV
15. Consider the following statement.
Statement I: 0.1M H_3PO_3 (aq) solution has normality equal to 0.3 N when completely reacted with NaOH.
Statement II: H_3PO_3 is tribasic acid
 (A) Both statement I and II are correct
 (B) Both statement I and II are incorrect
 (C) Ist statement is correct but II statement is incorrect
 (D) Ist statement is incorrect but II statement is correct
16. 5g sample contains only Na_2CO_3 and Na_2SO_4 . This sample is dissolved and the volume made up to 250 ml. 25 ml of this solution neutralizes 20 ml of 0.1 M H_2SO_4 . Calculate the % of Na_2SO_4 in the sample.
 (A) 42.4 (B) 78.8 (C) 57.6 (D) 21.2
17. The average molar mass of chlorine is 35.5 mol^{-1} . The ratio of ^{35}Cl to ^{37}Cl in naturally occurring chlorine is close to:
 (A) 2 : 1 (B) 3 : 1 (C) 4 : 1 (D) 1 : 1
18. The mass (in gms) of 112 mL of O_2 gas at STP is:
 (A) 32 g (B) 0.16 g (C) 16 g (D) 0.032 g
19. Percentage loss in mass, when $NaHCO_3(s)$ is heated in open vessel.
 (A) 36.9 (B) 25.4 (C) 74.9 (D) 52.3
20. A 0.13 g sample of a purified metal carbonate was dissolved in 50.0 mL of 0.1M HCl and boiled to eliminate CO_2 . Back titration of the excess HCl required 24.0 mL of 0.1 M NaOH. Identify the metal.
 (A) Ca (B) Na (C) Mg (D) Insufficient Data

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. 20 ml of HCl solution gives 2.87 g AgCl with an excess of AgNO_3 . What is the molarity of acid?
(Ag = 108, Cl = 35.5, N = 14, O = 16)
22. Ratio of masses of methane and ethane in a gas mixture is 4 : 5. The ratio of number of their molecules in the mixture is x : y. Value of $x \times y$ is:
23. On heating 1.06g of hydrated CaCl_2 i.e. $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$ to dryness, 0.8g of anhydrous salt remained. What is the value of x? (Mo. Ca = 40g, Cl = 35.5 g)
24. What volume of water (in litres) must be added to 0.4 litre of 10N H_2SO_4 solution to make it exactly 2N. (Report your answer to nearest integer)
25. 2.5L of 1M NaOH solution is mixed with 3L of 0.5M $\text{Ca}(\text{OH})_2$ solution. The concentration of the resultant OH^- in a solution is:

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If α, β are roots of $x^2 + 3x - 1 = 0$ then the equation whose roots are $\alpha^2 + 2$ and $\beta^2 + 2$ is:
 (A) $x^2 - x - 2 = 0$ (B) $x^2 + 15x - 27 = 0$
 (C) $x^2 - 15x + 27 = 0$ (D) $x^2 + 15x + 27 = 0$
- If x and y are distinct real numbers and $\log_2 x + \log_x 2 = \log_2 y + \log_y 2 = \frac{5}{2}$ then $x^2 + y^2$ is equal to:
 (A) 6 (B) $16 + \sqrt{2}$ (C) $4 + \sqrt{2}$ (D) 18
- If $\log_2(3x - 2) = \log_{1/2} x$ then:
 (A) $x = 1$ and $\frac{-1}{3}$ (B) $x = 1$ only (C) $x = -\frac{1}{3}$ only (D) No real value of x
- $\left| \frac{x+3}{x-2} \right| = 2$ then x is equal to:
 (A) $7, \frac{1}{3}$ (B) $\frac{1}{3}, -7$ (C) $-\frac{1}{3}, 7$ (D) $-\frac{1}{3}, -7$
- α, β, γ are roots of $x^3 - 7x^2 + 4x - 3 = 0$ then the equation whose roots are $\frac{1}{\alpha+1}, \frac{1}{\beta+1}$ and $\frac{1}{\gamma+1}$ is:
 (A) $15x^3 - 21x^2 + 10x + 1 = 0$ (B) $15x^3 - 21x^2 + 10x - 1 = 0$
 (C) $15x^3 + 21x^2 - 10x + 1 = 0$ (D) $15x^3 + 21x^2 - 10x - 1 = 0$
- The expression $(ax^2 + bx + c)$ is always positive for all real values of x then the sufficient condition is:
 (A) $a > 0$ (B) $b^2 - 4ac < 0$ (C) $a(b^2 - 4ac) < 0$ (D) $a > 0$ and $b^2 - 4ac < 0$
- $\frac{(x^3 - 1)(x + 1)^2}{(x - 3)^4} \geq 0$, then complete solution set for x is:
 (A) $[1, \infty)$ (B) $[1, \infty) \cup \{-1\}$ (C) $[1, \infty) - \{3\}$ (D) None of these
- If $y = x^2 - 10x + 7$ then y cannot be equal to:
 (A) 10 (B) 0 (C) -10 (D) -20
- If $\frac{\alpha+1}{\alpha-1} + \frac{\alpha-1}{\alpha+1} = K$ and $\frac{\beta+1}{\beta-1} + \frac{\beta-1}{\beta+1} = K$ then value of $\alpha\beta$ is equal to: ($\alpha \neq \beta$)
 (A) $\frac{K+2}{K-1}$ (B) $-\frac{(K+2)}{K-2}$ (C) $\frac{K-2}{K+2}$ (D) None of these
- If $12(9^x) - 35(6^x) + 18(4^x) = 0$ then sum of real values of x satisfying the given equation is:
 (A) 0 (B) $\frac{35}{12}$ (C) 1 (D) None of these

11. $f(x) = ax^2 + bx + c$ and $f(\alpha)f(\beta) < 0$ where a, b, c, α and β are all real numbers and following statements are:
 I: $f(x) = 0$ has two distinct real roots
 II: $f(x) = 0$ has exactly one value of x in the interval (α, β)
 III: $f(x) = 0$ may or may not have a real root.
 Correct statement(s) is/are:
 (A) I only (B) III only (C) Both I and II (D) II only
12. If both the roots of the equation $x^2 - 3kx + 4 = 0$ are less than 2 then:
 (A) $k < \frac{4}{3}$ (B) $k \leq -\frac{4}{3}$ (C) $-\frac{4}{3} \leq k \leq \frac{4}{3}$ (D) $k > \frac{4}{3}$
13. If '1' is between distinct roots of $kx^2 - 2x + k = 0$ then:
 (A) $0 < k < 1$ (B) $-1 < k < 1$ (C) $k > -1$ (D) $k \in (-1, 1) - \{0\}$
14. If $\frac{x^2 - |x| - 6}{(x + 4)} \geq 0$ then solution set of values of x is:
 (A) $(-\infty, -4) \cup [-3, 3]$ (B) $(-\infty, -2] \cup [-1, \infty)$
 (C) $(-4, -3] \cup [3, \infty)$ (D) $(-4, 0) \cup (0, 3)$
15. If α and β are roots of $x^2 + \lambda x + (\lambda + 1) = 0$ then minimum value of $\alpha^2 + \beta^2$ occurs at λ equal to:
 (A) 1 (B) -1 (C) 2 (D) -2
16. The roots of the equation $x^3 + kx^2 + 1 = 0$ are in arithmetic progression then the value of ' k ' is:
 (A) $\frac{3}{2^{1/3}}$ (B) $\frac{-3}{2^{1/3}}$ (C) -3 (D) $\frac{-3}{2}$
17. $\frac{(x^2 - 1)(x^2 - 4)(x^2 - 9)}{(x^2 - 16)(x^2 - 25)} \leq 0$ then the sum of positive integer values of x is equal to:
 (A) 15 (B) 10 (C) 9 (D) 6
18. If x takes all real values then range of $f(x) = \frac{x^2 - 2x + 4}{x^2 + 2x + 4}$ is:
 (A) $[3, \infty)$ (B) $(-\infty, \frac{1}{3}]$ (C) $(-\infty, \frac{1}{3}] \cup [3, \infty)$ (D) $[\frac{1}{3}, 3]$
19. Minimum value of $x^2 + ax + b$ is zero at $x = -2$ then ordered pair (a, b) is:
 (A) (4, 3) (B) (3, 4) (C) (4, 4) (D) (-4, 4)
20. $y = (x - 1)(x - 2) + (x - 2)(x - 3) + (x - 3)(x - 1)$ then roots of the equation $y = 0$ are:
 (A) Real and equal (B) Real, distinct and rational
 (C) Real, distinct and irrational (D) None of these

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. The sum of integer values of x for the given inequation $1 \leq |x - 2| \leq 3$ is $3k$ then k is equal to:
22. The number of real roots of the equation $2x^2 - 7|x| + 3 = 0$ is equal to:
23. The number of real roots of the equation $x^4 + x^3 + x^2 + 2 = 0$ is:
24. If $5 + \sqrt{2}$ is a root of the equation $x^2 + ax + b = 0$ where a and b are both rational numbers then the value of $(2a + b)$ is equal to:
25. If the equations $x^2 - Kx + 5 = 0$ and $x^2 - (K + 1)x + 10 = 0$ have exactly one common root then value of K is equal to:

SPACE FOR ROUGH WORK

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